

MADSON ALLAN DE LUNA ARAGÃO

Bioinformatics Engineer | Software Development | Data Analyst

Belo Horizonte, MG, Brazil | madsondeluna@gmail.com

madsondeluna.com | GitHub | LinkedIn | Google Scholar | ORCID

PROFESSIONAL SUMMARY

Bioinformatics engineer and scientific software developer with approximately 10 years of research experience and over 5 years in product development and product, project and team management. Builds web applications, APIs, Python tools and pipelines that integrate biological data, machine learning and high-performance computing (HPC). Developer of AMPIdentifier, GenVar and BILBO, with experience in genomics, structural biology and generative-model research for antimicrobial peptides. Background in AI and IoT products, requirements, MVPs, roadmaps and stakeholder management. Currently Bioinformatics Engineer and Head of Data Science at Biotech Business; Nextflow Ambassador.

PROFESSIONAL EXPERIENCE

Biotech Business

Oct 2025 – Present

Bioinformatics Engineer & Head of Data Science | Aug 2026 – Present

Mid-level Bioinformatician | Oct 2025 – Jul 2026

- Develop computational solutions for biological data analysis in agribusiness, including machine learning and computer vision applications, APIs and database integrations.
- Translate scientific and business needs into requirements, priorities and software deliverables; contribute to roadmaps and solution development with research and business teams.

PickCells | Research and Development Analyst

Oct 2020 – Aug 2023 | Recife, Brazil

- Prototyped, developed and validated IoT and AI products for pattern recognition in clinical images; defined requirements, MVPs, roadmaps and KPIs.
- Conducted market and user research, collected feedback and aligned priorities with stakeholders; organized delivery using Agile, Lean and Scrum practices.

Hospital das Clínicas, UFPE / EBSEERH | Clinical Pathology Intern

Oct 2021 – Mar 2022 | Recife, Brazil

- Supported sample screening, hematology, biochemistry, urinalysis, microbiology, hormone assays and serology; performed laboratory analyses and quality control procedures.

PickCells | Research, Development and Innovation Intern

Apr 2020 – Aug 2020 | Recife, Brazil

- Contributed to the development of solutions for molecular diagnosis of SARS-CoV-2.

TECHNICAL AND PRODUCT SKILLS

Programming and software: Python, R, Bash/Shell, SQL, Java, JavaScript, HTML/CSS; FastAPI, Flask, React, REST APIs, GraphQL and CLI development.

Data and visualization: pandas, NumPy, applied statistics, ETL, data modeling and governance; PostgreSQL, MySQL, SQLite, Redis; Tableau and Power BI.

AI and machine learning: scikit-learn, Random Forest, SVM, gradient boosting, feature engineering, model validation and computer vision; research using deep learning, Transformers and generative models.

Workflows and HPC: Linux/Unix, Slurm, Nextflow, Snakemake, automation and resource management; Docker, Docker Compose, Git/GitHub, pytest, documentation and release management.

Bioinformatics and modeling: genomics, transcriptomics, genome assembly, mining and annotation; variant analysis, protein and membrane modeling, protein design, docking and molecular dynamics; GROMACS and PyMOL.

Products and collaboration: product lifecycle, requirements, MVPs, roadmaps, KPIs, backlog, Scrum, Kanban, Lean, UCD and Design Thinking; team leadership, stakeholder management, user research and technical/business communication.

Work tools: Figma, Miro, Jira, Trello, Azure, Notion and Google Workspace; prototyping, wireframing, customer journeys and data storytelling.

SELECTED SOFTWARE PROJECTS

AMPIdentifier | Python, scikit-learn, CLI

Antimicrobial peptide prediction tool integrating FASTA input, physicochemical descriptors, classification and result export. Available as a documented web application and Python package. INPI software registration: BR 51 2025 005859-4.

GenVar Dashboard | FastAPI, React, Redis, Docker, pytest

Gene and variant exploration application integrating Ensembl, gnomAD, ClinVar, AlphaFold, UniProt and MyVariant.info. Front-end, API and cache architecture, containerization and service/integration tests. Software Engineering MBA project.

BILBO | Python, CLI, GROMACS

Lipid bilayer construction and protein/peptide positioning tool; generates initial structures and reports to prepare molecular dynamics simulations. Includes documentation and automated tests.

GETVar | Python, Flask, Snakemake, Bootstrap

VCF processing and genetic variant annotation application, with workflows integrating dbSNP, ClinVar and Ensembl.

Plant eIF4E Atlas | Python, SQL, ETL, APIs

Plant eIF4E data integration project featuring an ETL pipeline, data model and taxonomic/functional exploration. Developed during the Data Science and Analytics specialization.

More projects and applications: github.com/madsondeluna | Full portfolio

APPLIED RESEARCH EXPERIENCE

UFMG | Bioinformatics Researcher, doctoral research

Aug 2024 – Present | Belo Horizonte, Brazil

- Research deep learning and generative models for antimicrobial peptide identification and design; integrate sequence and molecular-property analyses, structural modeling and simulations of membrane interactions.

UFPE, Department of Genetics | Bioinformatics Researcher, master's research

Jun 2022 – Jul 2024 | Recife, Brazil

- Developed AMPIdentifier, including data preparation, descriptors and ML model comparison. Identified and characterized five *Ricinus communis* defensins using genomics, transcriptomics and molecular modeling.

UFPE, Plant Genetics and Biotechnology Laboratory | Undergraduate Researcher

Jan 2020 – May 2022 | Recife, Brazil

- Investigated bioinspired antimicrobial peptides and *Vigna eIF4E* sequences associated with defense against pathogens.

Fiocruz, Aggeu Magalhães Institute | Undergraduate Researcher

Nov 2016 – Dec 2019 | Recife, Brazil

- Worked on structural biology, computational chemistry, protein modeling and engineering for diagnostic and vaccine-related research in Virology and Experimental Therapy.

UFPE, Keizo Asami Institute | Undergraduate Researcher

May 2015 – Dec 2016 | Recife, Brazil

- Investigated molecular biology, human genetics and bioinformatics, including ancestry markers and phenotype prediction for forensic genetics in collaboration with Pernambuco's forensic science authority.

EDUCATION

PhD in Bioinformatics | Federal University of Minas Gerais (UFMG) | 2024 – Present; in progress

MBA in Software Engineering | University of São Paulo (USP) | Expected completion: December 2026

Postgraduate Specialization in Data Science and Analytics | Pontifical Catholic University of Rio de Janeiro (PUC-Rio) | 2026

MSc in Genetics and Molecular Biology | Federal University of Pernambuco (UFPE) | 2024

BSc in Biomedical Sciences | UFPE | 2022

Technical Diploma in Software Development | ETEPE | 2013

Languages: Portuguese: native | English: advanced

SELECTED CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

Docker Foundations Professional Certificate | Docker, Inc. | 2026

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | Oracle | 2025

Nextflow Training | Seqera | 2025

Slurm Workload Manager | National Laboratory for Scientific Computing, Brazil

Microsoft Azure AI Essentials: Workloads and Machine Learning | 2025

Career Essentials in GitHub Professional Certificate | 2024

Machine Learning and Analytics | CCEC / PUC-Rio | 2025

Generative AI | MIT Professional Education | 2025

Advanced Gemini for Developers | 2024

Requirements Engineering and Agile Product Management | PUC-Rio | 2025

Agile Project Management Professional Certificate | Atlassian | 2025

Google | 2026: Data Responsibility; Stakeholder Management; Make Data-Driven Decisions; Data Analytics and Analytical Thinking.

RECOGNITION AND COMMUNITY

Nextflow Ambassador | Nextflow / Seqera | Jan 2026 – Present. Member of the ambassador program, sharing knowledge on reproducible workflows and supporting Nextflow adoption within the bioinformatics community.

Next Generation Bioinformatician | AB3C, X-Meeting 2025.

Editor's Choice, Viruses | Coauthored paper on cowpea resistance to potyvirus, 2025.

Travel Grant | School on Biological Physics and Biomolecular Simulations in the Machine Learning Era, ICTP-SAIFR, 2025.

Travel Grant | AI for Protein Design (AI4PD), The Protein Society.

Young Geneticist Award of the Northeast | XXI Northeast Genetics Meeting.

Best Poster Award | XIII Genetics and Molecular Biology Program Meeting.

Certificate of Excellence in Peer Reviewing | Elsevier.

Ranked first in admission to the MSc in Genetics and Molecular Biology (UFPE) and PhD in Bioinformatics (UFMG).

Honorable mentions: Human and Forensic Genetics (CNPq/UFPE); Postgraduate Genetics Meeting (UFPE); Pernambuco Chemistry Olympiad (OPEQ).

TEACHING AND SCIENTIFIC COMMUNICATION

Invited instructor: Molecular Docking & Dynamics: Breathing Motion into Life's Building Blocks, 1st and 2nd editions (2025).

Invited instructor: Molecular Modeling with Machine Learning Techniques, 1st and 2nd editions (2025).

Invited instructor: Bioinformatics: A Theoretical-Practical Approach (2024).

Invited instructor: Introduction to Bioinformatics: From DNA to Proteins, covering databases, annotation and protein modeling, UNICAP (2022).

Teaching assistant: Human Genetics; Molecular Tools Applied to Clinical Diagnosis, UFPE (2017–2018). Developed PyMOL teaching material for Basic Bioinformatics (2026).

PUBLICATIONS

The Regulatory Army of Plant Defense: Transcription Factors in the War for Plant Immunity. International Journal of Molecular Sciences, 27, 7315 (2026).

Unveiling Three Functionally Diverse Isoforms of eIF4E in Cowpea Through a Multi-Omics Approach. Agronomy, 16, 766 (2026).

Transposable elements: Functional aspects and applications as drivers of crop innovation. Crop Science, 66, e70257 (2026).

Deciphering Cowpea Resistance to Potyvirus: Assessment of eIF4E Gene Mutations and Their Impact on the eIF4E-VPg Protein Interaction. Viruses, 17, 1050 (2025).

Omics-driven bioinformatics for plant lectins discovery and functional annotation — A comprehensive review. International Journal of Biological Macromolecules, 135511 (2024).

Association strength of E6 to E6AP/p53 complex correlates with HPV-mediated oncogenesis risk. Biopolymers, 113, e23524 (2022).

Approaches for Identification and Validation of Antimicrobial Compounds of Plant Origin: A Long Way from the Field to the Market. Chapter in Eco-Friendly Biobased Products Used in Microbial Diseases, CRC Press (2022).